

The Eschede Train derailment

Taken directly from: https://en.wikipedia.org/wiki/Eschede_derailment

"The **Eschede derailment** occurred on 3 June 1998, near the village of Eschede in the Celle district of Lower Saxony, Germany, when a high-speed train derailed and crashed into a road bridge. 101 people were killed and 88 were injured. It remains the worst rail disaster in the history of the FRG and the worst high-speed-rail disaster worldwide. The cause was a single fatigue crack in one wheel that, when it failed, resulted in a part of the wheel becoming caught in a set of points, effectively changing the setting of the points whilst the train was passing over it. This led to the train's carriages going down two separate tracks, causing the train to derail and crash into the pillars of a concrete road bridge, which then collapsed and crushed two coaches. The remaining coaches and the rear power car crashed into the wreckage.

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Causes

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Wheel design

The ICE 1 trains were originally equipped with single-cast wheelsets, known as Monobloc wheels. Once in service it soon became apparent that this design could, as a result of metal fatigue and out-of-round conditions, result in resonance and vibration at cruising speed. Passengers noticed this particularly in the restaurant car, where there were reports of loud vibrations in the dinnerware and of glasses "creeping" across tables.

Managers in the railway organisation had experienced these severe vibrations on a trip and asked to have the problem solved. In response engineers decided that to solve the problem, the suspension of ICE cars could be improved with the use of a rubber damping ring between the rail-contacting steel tire and the steel wheel body. A similar design had been employed successfully in trams around the world (known as resilient wheels), at significantly lower speeds. This kind of wheel, dubbed a "wheel-tyre" design, consisted of a wheel body surrounded by a 20-millimetre (0.79 in) thick rubber damper and then a relatively thin metal tyre. The new design was not tested at high speed in Germany before it was made operational, but was successful at resolving the issue of vibration at cruising speeds. Decade-long experience at high speed gathered by train manufacturers and railway companies in other European countries and Japan was not considered.

At the time, no facilities existed in Germany that could test the actual failure limit of the wheels, and so complete prototypes were never tested physically. The design and specification relied greatly on available materials data and theory. The very few laboratory and rail tests that were performed did not measure wheel behaviour with extended wear conditions or speeds greater than normal cruising. Nevertheless, over a period of years the wheels proved themselves apparently reliable and, until the accident, had not caused any major problems.

In July 1997, nearly one year before the disaster, Üstra the company that operates Hanover's tram network, discovered fatigue cracks in dual block wheels on trams running at about 24 km/h (15 mph). It began changing wheels before fatigue cracks could develop, much earlier than was legally required by the specification. Üstra reported its findings in a warning to all other users of wheels built with similar designs, including Deutsche Bahn, in late 1997. According to Üstra, Deutsche Bahn replied by stating that they had not noticed problems in their trains.

The Fraunhofer Institute for Structural Durability and System Reliability (LBF) in Darmstadt was charged with the task of determining the cause of the accident. It was revealed later that the institute had told the DB management as early as 1992 about its concerns about possible wheel-tyre failure.

It was soon apparent that dynamic repetitive forces had not been accounted for in the statistical failure modelling done during the design phase, and the resulting design lacked an adequate margin of safety. The following factors, overlooked during design, were noted:

1. The tyres were flattened into an ellipse as the wheel turned through each revolution (approximately 500,000 times during a typical day in service on an ICE train), with corresponding fatigue effects.
2. In contrast to the Monobloc Wheel design, cracks could also form on the inside of the tyre.
3. As the tyre became thinner due to wear, the dynamic forces were exaggerated, resulting in crack growth.
4. Flat spots and ridges or swells in the tyre dramatically increased the dynamic forces on the assembly and greatly accelerated wear.

Failure to stop train

Failing to stop the train resulted in a catastrophic series of events. Had the train been stopped immediately after the disintegration of the wheel, it is unlikely that the subsequent events would have occurred.

Valuable time was lost when the train manager refused to stop the train until he had investigated the problem himself, saying this was company policy. This decision was upheld in court, absolving the train manager of all charges. Given that he was a customer service employee and not a train maintainer or engineer, he had no more authority to make an engineering judgment about whether or not to stop the train than the passenger anyway.

Maintenance

About the time of the disaster, the technicians at Deutsche Bahn's maintenance facility in Munich used only standard flashlights for visual inspection of the tyres, instead of metal fatigue detection equipment. Previously, advanced testing machines had been used; as the equipment generated many false positive error messages, however, it was considered unreliable and its use was discontinued. During the week prior to the Eschede disaster, three separate automated checks indicated that a wheel was defective. Investigators discovered, from a maintenance report generated by the train's on-board computer, that two months prior to the Eschede disaster, conductors and other train staff filed eight separate complaints about the noises and vibrations generated from the bogie with the defective wheel; the company did not replace the wheel. Deutsche Bahn said that its inspections were proper at the time and that the engineers could not have predicted the wheel fracture.

Other factors

The design of the overbridge may have also contributed to the accident because it had two thin piers holding up the bridge on either side, instead of the spans going from solid abutments to solid abutments. The Granville Train Disaster of 1977 had a similar weakness in its bridge. The bridge built after the disaster is a cantilvered design and does not have this vulnerability.

Another contributing factor to the casualty rate was the use of welds in the carriage bodies that "un-zipped" during the crash."